

EEE283

Digital Logic Design

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Topic 5:
Combinational Logic Circuits for Arithmetic Operations

Addition of Unsigned Number

x	0	0	1	1
$+ y$	$+ 0$	$+ 1$	$+ 0$	$+ 1$
$c \ s$	0 0	0 1	0 1	1 0

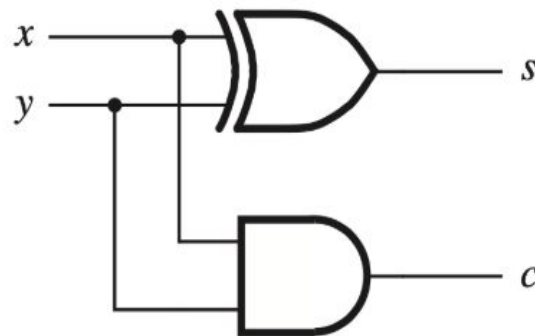
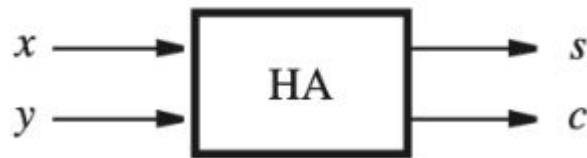
Carry $\xrightarrow{\quad}$ $\uparrow$ $\uparrow$ $\xleftarrow{\quad}$ Sum

Half Adder

		Carry	Sum
x	y	c	s
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$$s = \bar{x}y + x\bar{y} = x \oplus y$$

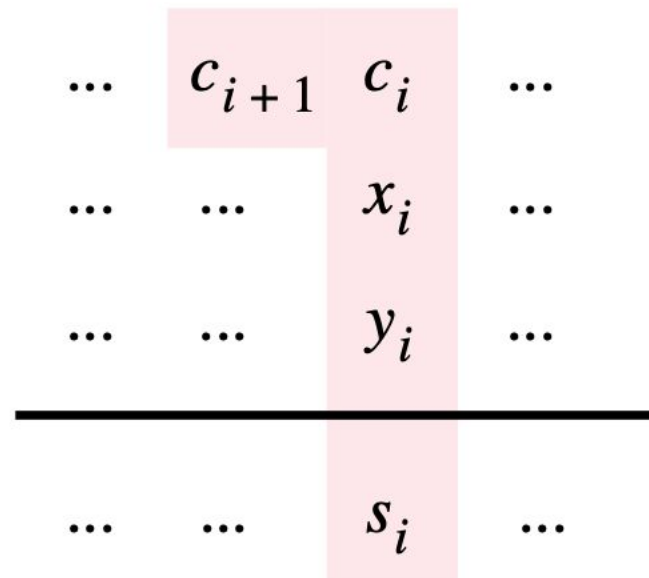
$$c = xy$$



N-bit Binary Addition

Generated carries $\longrightarrow$ 1 1 1 0

$$\begin{array}{r}
 X = x_4 x_3 x_2 x_1 x_0 \quad 0\ 1\ 1\ 1\ 1 \quad (15)_{10} \\
 + Y = y_4 y_3 y_2 y_1 y_0 \quad +\ 0\ 1\ 0\ 1\ 0 \quad +\ (10)_{10} \\
 \hline
 S = s_4 s_3 s_2 s_1 s_0 \quad 1\ 1\ 0\ 0\ 1 \quad (25)_{10}
 \end{array}$$



Full Adder

c_i	x_i	y_i	c_{i+1}	s_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$x_i y_i$					
c_i		00	01	11	10
0			1		1
1		1		1	

$$s_i = \bar{x}_i y_i \bar{c}_i + x_i \bar{y}_i \bar{c}_i + \bar{x}_i \bar{y}_i c_i + x_i y_i c_i$$

$$s_i = (\bar{x}_i y_i + x_i \bar{y}_i) \bar{c}_i + (\bar{x}_i \bar{y}_i + x_i y_i) c_i$$

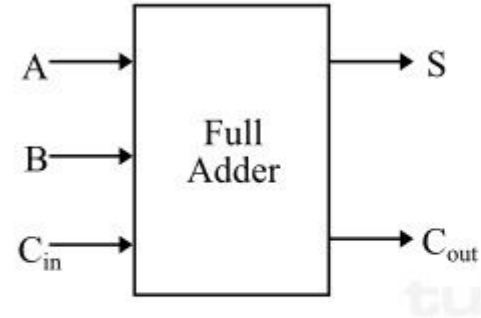
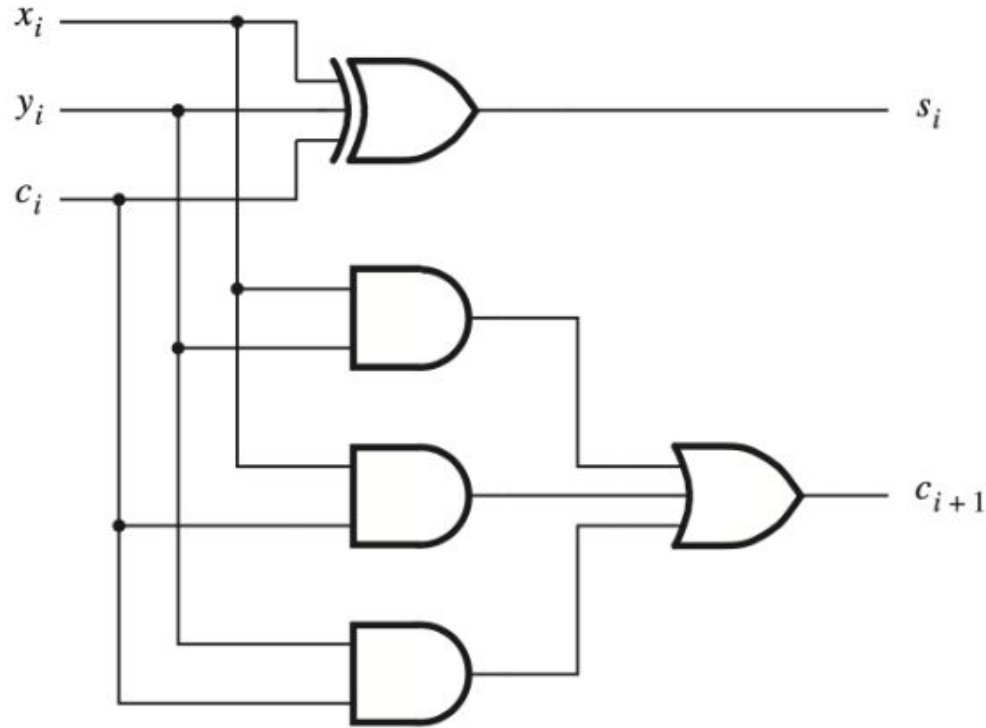
$$= (x_i \oplus y_i) \bar{c}_i + \overline{(x_i \oplus y_i)} c_i$$

$$= (x_i \oplus y_i) \oplus c_i$$

$x_i y_i$					
c_i		00	01	11	10
0				1	
1			1	1	1

$$c_{i+1} = xy + xc_i + yc_i$$

Full Adder Logic Diagram



Full Adder using Half Adders

Half Adder

$$S_{HA1} = x_i \oplus y_i$$

$$C_{HA1} = x_i y_i$$

Full Adder

$$S_i = (x_i \oplus y_i) \oplus C_i$$

$$C_{i+1} = (x_i C_i) + (y_i C_i) + (x_i y_i)$$

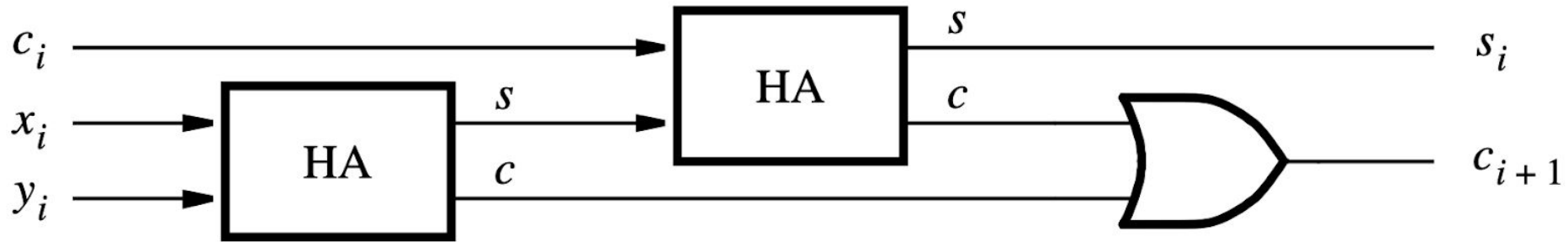
Cascading 2 Half -Adders

$$S_{HA2} = S_{HA1} \oplus C_i$$

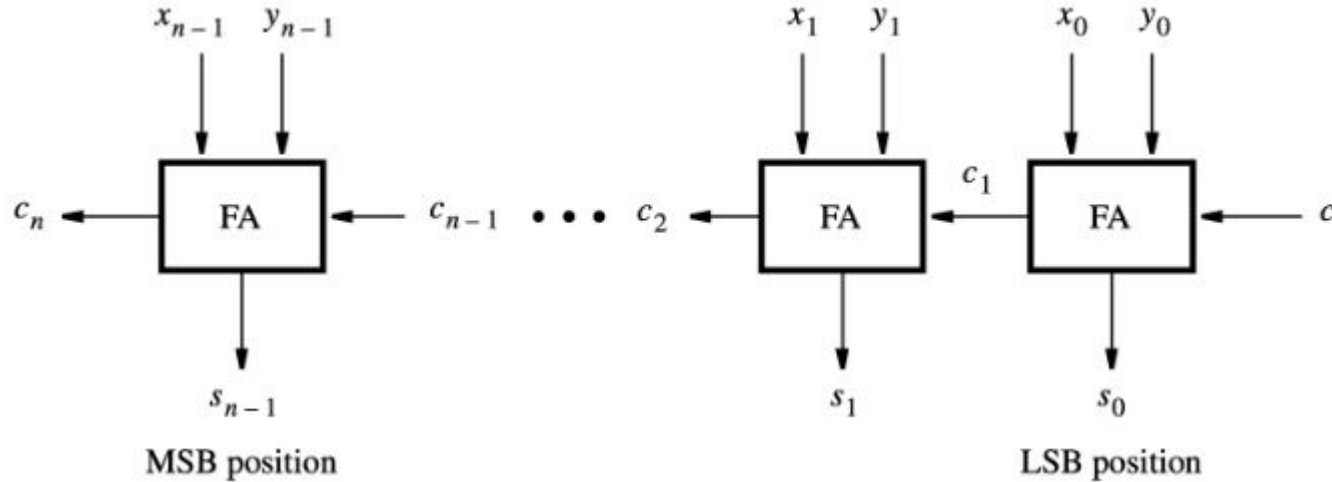
$$\begin{aligned} C_{HA2} &= S_{HA1} C_i \\ &= (x_i \oplus y_i) C_i \\ &= (x_i \bar{y}_i + \bar{x}_i y_i) C_i \end{aligned}$$

$$\begin{aligned} C_{HA2} + C_{HA1} &= C_i (x_i \bar{y}_i + \bar{x}_i y_i) + x_i y_i \\ &= C_i x_i \bar{y}_i + C_i \bar{x}_i y_i + x_i y_i + x_i y_i \\ &= x_i y_i + C_i x_i \bar{y}_i + x_i y_i + C_i \bar{x}_i y_i \\ &= x_i (y_i + C_i \bar{y}_i) + y_i (x_i + C_i \bar{x}_i) \\ &= x_i (y_i + C_i) + y_i (x_i + C_i) \\ &= x_i y_i + x_i C_i + y_i C_i \end{aligned}$$

Full Adder using Half Adders

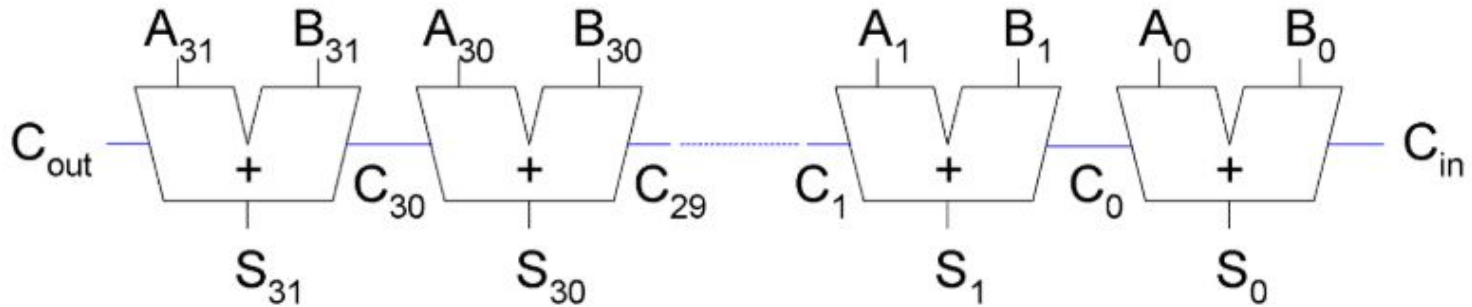


N-bit Ripple Carry Adder



N-bit Ripple Carry Adder

- Chain 1-bit adders together
- Carry ripples through entire chain
- Disadvantage: **slow**



$$t_{\text{ripple}} = N t_{FA}$$

Performance Issues

- Each carry calculation needs 2 cascaded gates.
- Gate delay for each gate Δt
- Final result of the addition will be valid after a delay of $n\Delta t$,

Signed Numbers

$$\begin{array}{r} (+5) \\ - (+2) \\ \hline (+3) \end{array}$$

$$\begin{array}{r} 0101 \\ - 0010 \\ \hline \end{array}$$



$$\begin{array}{r} 0101 \\ + 1110 \\ \hline 10011 \end{array}$$

↑
ignore

$$\begin{array}{r} (+5) \\ - (-2) \\ \hline (+7) \end{array}$$

$$\begin{array}{r} 0101 \\ - 1110 \\ \hline \end{array}$$



$$\begin{array}{r} 0101 \\ + 0010 \\ \hline 0111 \end{array}$$

$$\begin{array}{r} (-5) \\ - (+2) \\ \hline (-7) \end{array}$$

$$\begin{array}{r} 1011 \\ - 0010 \\ \hline \end{array}$$



$$\begin{array}{r} 1011 \\ + 1110 \\ \hline 11001 \end{array}$$

↑
ignore

$$\begin{array}{r} (-5) \\ - (-2) \\ \hline (-3) \end{array}$$

$$\begin{array}{r} 1011 \\ - 1110 \\ \hline \end{array}$$



$$\begin{array}{r} 1011 \\ + 0010 \\ \hline 1101 \end{array}$$

Arithmetic Overflow

- If n bits are used to represent signed numbers, then the result must be in the range -2^{n-1} to $2^{n-1} - 1$.
- If the result does not fit in this range, then we say that *arithmetic overflow* has occurred.
- To ensure the correct operation of an arithmetic circuit, it is important to be able to detect the occurrence of overflow.

Arithmetic Overflow

$$\begin{array}{r}
 (+7) \quad 0111 \\
 + (+2) \quad + 0010 \\
 \hline
 (+9) \quad 1001 \\
 \quad c_4 = 0 \\
 \quad c_3 = 1
 \end{array}$$

$$\begin{array}{r}
 (-7) \quad 1001 \\
 + (+2) \quad + 0010 \\
 \hline
 (-5) \quad 1011 \\
 \quad c_4 = 0 \\
 \quad c_3 = 0
 \end{array}$$

$$\begin{array}{r}
 (+7) \quad 0111 \\
 + (-2) \quad + 1110 \\
 \hline
 (+5) \quad 10101 \\
 \quad c_4 = 1 \\
 \quad c_3 = 1
 \end{array}$$

$$\begin{array}{r}
 (-7) \quad 1001 \\
 + (-2) \quad + 1110 \\
 \hline
 (-9) \quad 10111 \\
 \quad c_4 = 1 \\
 \quad c_3 = 0
 \end{array}$$

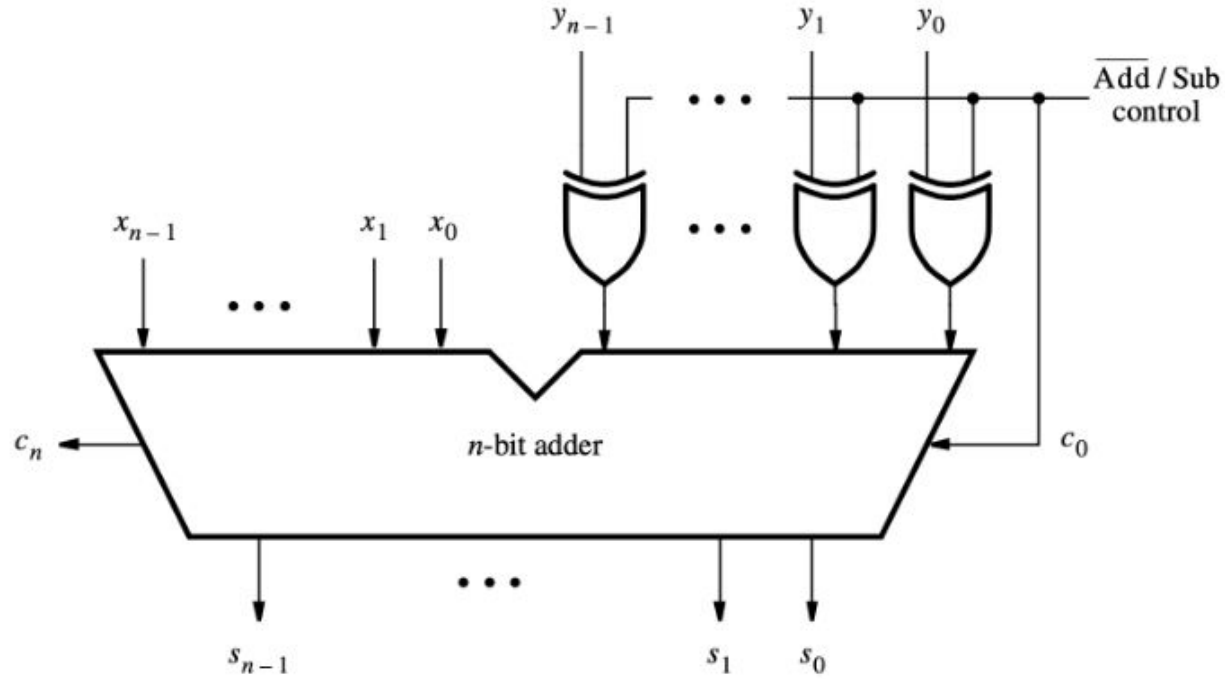
$$\begin{aligned}
 \text{Overflow} &= c_3 \bar{c}_4 + \bar{c}_3 c_4 \\
 &= c_3 \oplus c_4
 \end{aligned}$$

For N-bit operation

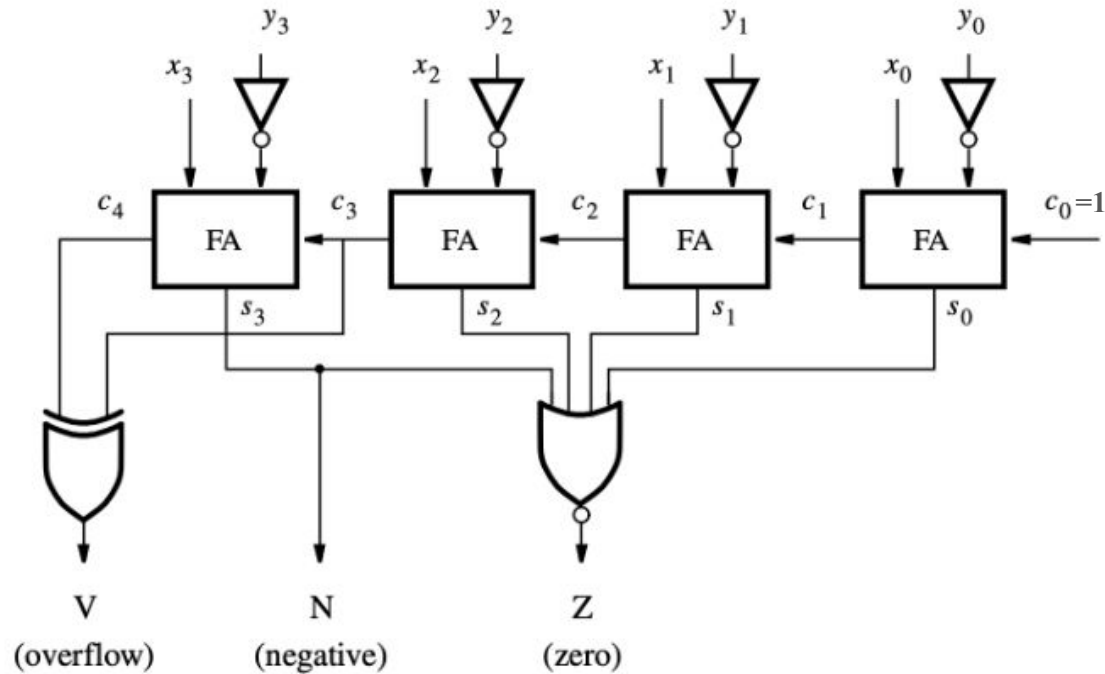
$$\textit{overflow} = x_{n-1}y_{n-1}\bar{s}_{n-1} + \bar{x}_{n-1}\bar{y}_{n-1}s_{n-1}$$

$$\textit{carryout} = x_{n-1}y_{n-1} + x_{n-1}\bar{s}_{n-1} + y_{n-1}\bar{s}_{n-1}$$

N- bit Subtraction



N- bit Comparator using Adders

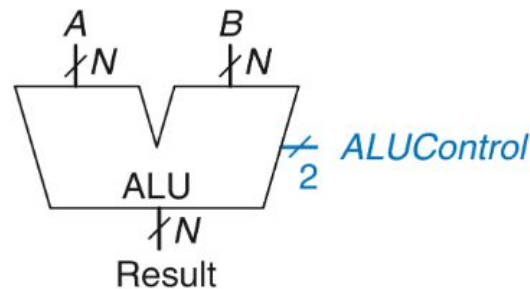


Arithmetic Logic Unit (ALU)

ALU should perform:

- Addition
- Subtraction
- AND
- OR

ALUControl _{1:0}	Function
00	Add
01	Subtract
10	AND
11	OR



Example: Perform $A + B$

$ALUControl = 00$

$Result = A + B$

Arithmetic Logic Unit (ALU)

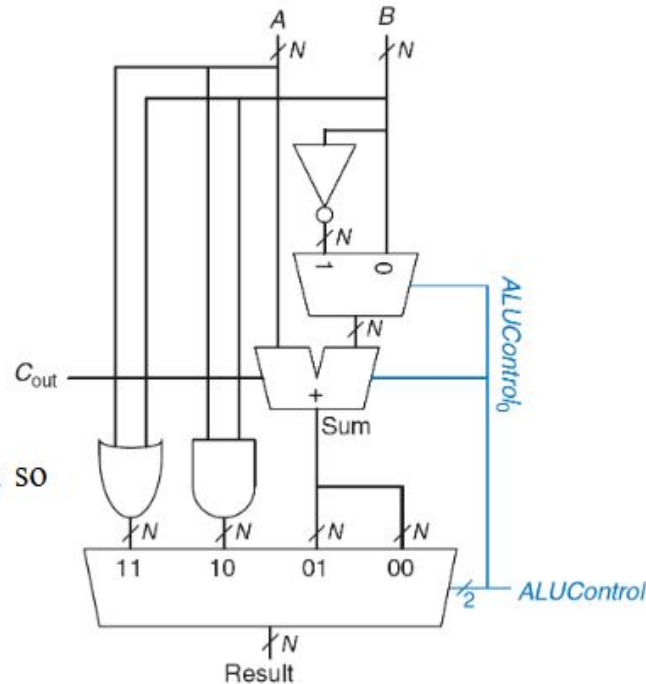
ALUControl _{1:0}	Function
00	Add
01	Subtract
10	AND
11	OR

Example: Perform $A \text{ OR } B$

$ALUControl_{1:0} = 11$

Mux selects output of OR gate as *Result*, so

$Result = A \text{ OR } B$



Arithmetic Logic Unit (ALU)

ALUControl _{1:0}	Function
00	Add
01	Subtract
10	AND
11	OR

Example: Perform $A + B$

$ALUControl_{1:0} = 00$

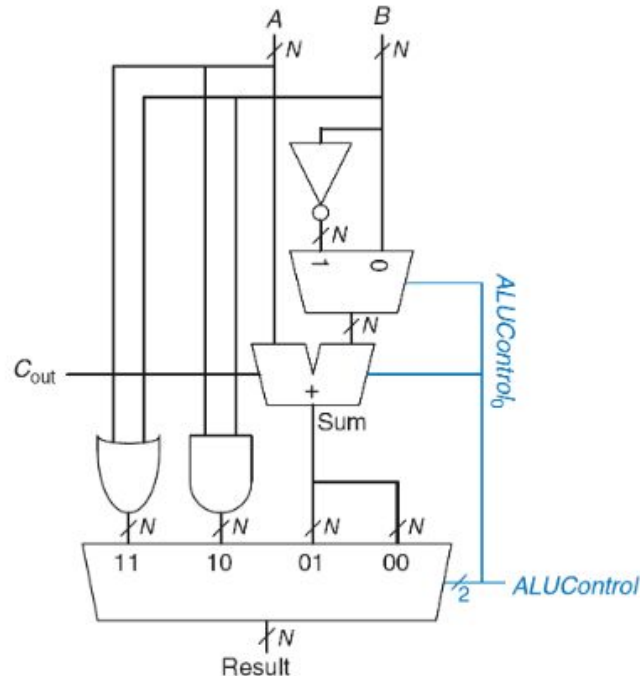
$ALUControl_0 = 0$, so:

C_{in} to adder = 0

2nd input to adder is B

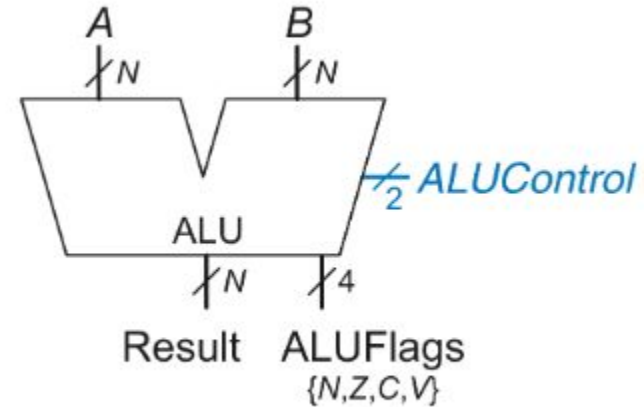
Mux selects Sum as $Result$, so

$Result = A + B$

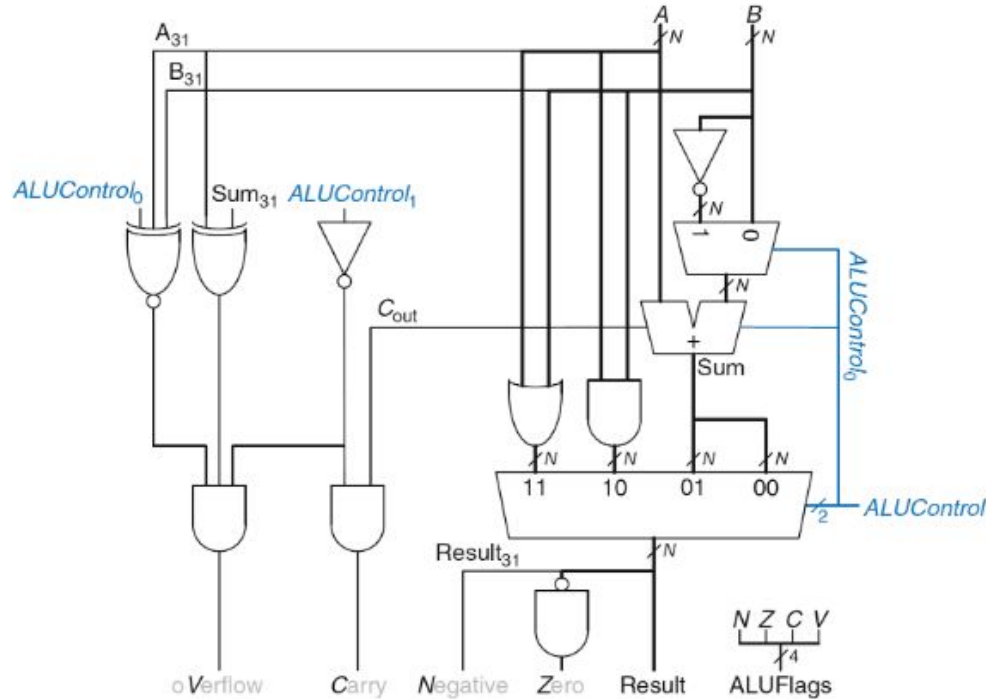


ALU with Status Flags

Flag	Description
<i>N</i>	Result is N egative
<i>Z</i>	Result is Z ero
<i>C</i>	Adder produces C arry out
<i>V</i>	Adder o V erflowed

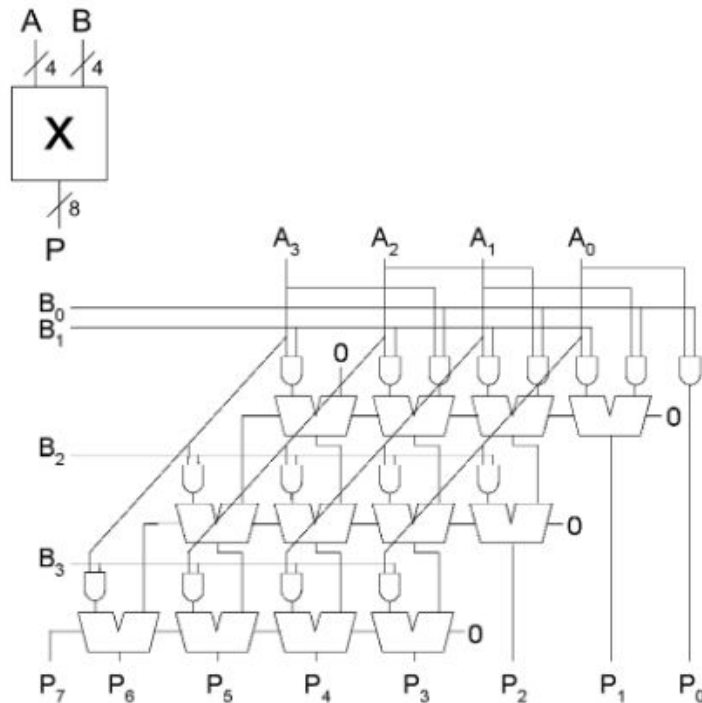


ALU with Status Flags

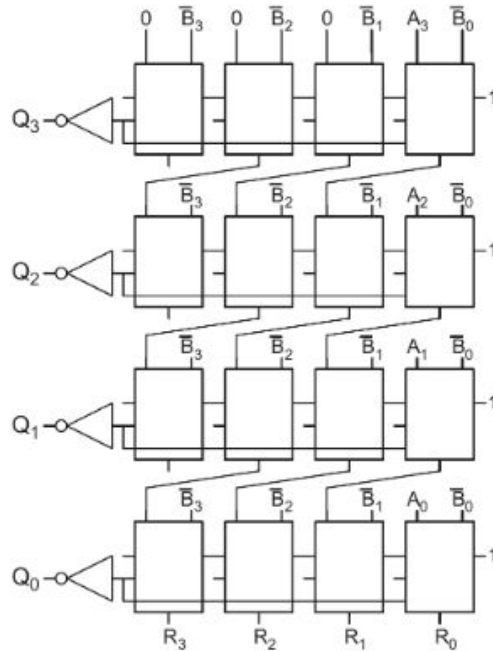


4x4 Multiplier

$$\begin{array}{r}
 \begin{array}{cccc}
 & A_3 & A_2 & A_1 & A_0 \\
 \times & B_3 & B_2 & B_1 & B_0 \\
 \hline
 & A_3B_0 & A_2B_0 & A_1B_0 & A_0B_0 \\
 & A_3B_1 & A_2B_1 & A_1B_1 & A_0B_1 \\
 & A_3B_2 & A_2B_2 & A_1B_2 & A_0B_2 \\
 + & A_3B_3 & A_2B_3 & A_1B_3 & A_0B_3 \\
 \hline
 P_7 & P_6 & P_5 & P_4 & P_3 & P_2 & P_1 & P_0
 \end{array}
 \end{array}$$

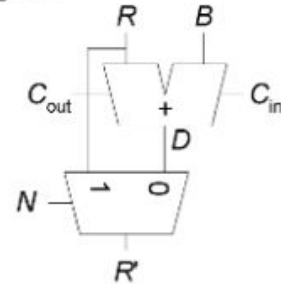


4x4 Divider



Each row computes one iteration of the division algorithm.

Legend



Division: $A/B = Q + R/B$

$R' = 0$

for $i = N-1$ to 0

$R = \{R' \ll 1, A_i\}$

$D = R - B$

if $D < 0$, $Q_i = 0$, $R' = R$

else $Q_i = 1$, $R' = D$

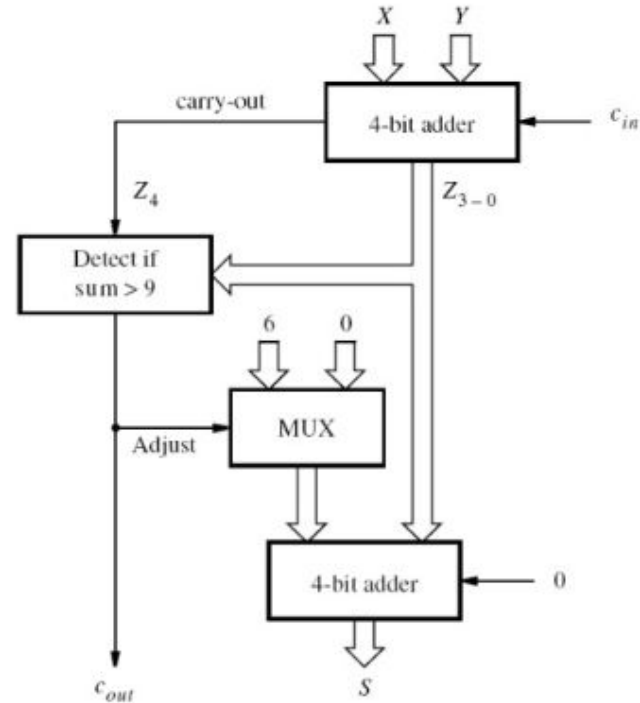
$R = R'$

BCD Adder

$$Z = X + Y$$

If $Z \leq 9$, then $S = Z$ and carry-out = 0

if $Z > 9$, then $S = Z + 6$ and carry-out = 1



BCD Adder (Compact)

